

1 Which statement is correct?

A One mole of a compound is the amount that contains the same number of atoms as there are in 12.0 g of carbon-12.

B The relative isotopic mass of beryllium-9 is given by the following expression.

$$\frac{\text{average mass of all isotopes of beryllium}}{\frac{1}{12} \text{ the mass of one atom of } ^{12}\text{C}}$$

C The relative atomic mass of nitrogen is given by the following expression.

$$\frac{\text{average mass of one atom of nitrogen}}{\frac{1}{12} \text{ the mass of one atom of } ^{12}\text{C}}$$

D The relative molecular mass of Q is given by the following expression.

$$\frac{\text{average mass of one atom of Q}}{\frac{1}{12} \text{ the mass of one atom of } ^{12}\text{C}}$$

2 10 cm³ of a gaseous hydrocarbon, C₂H₄, was exploded with an excess of oxygen. There